

LAN Architectures

- Topics covered:
 - 2-Tier and 3-Tier LAN Architecture
 - Spine-Leaf Architecture (Data Centre)
 - SOHO (Small Office/Home Office)

LAN Architectures & Network Design

Introduction

In previous topics, you studied technologies such as:

1. Routing and Switching
2. STP (Spanning Tree Protocol)
3. EtherChannel
4. OSPF
5. FHRPs
6. Switch Security Features

Now, the focus shifts to **network design and architecture**.

Network design is a broad field that requires understanding both networking principles and customer requirements. While best practices exist, there is rarely a single correct design for every network. The appropriate design depends on the needs of the organization.

Although network design is usually handled by experienced engineers, understanding basic architecture is important for configuring and troubleshooting enterprise networks.

This section covers:

1. Two-Tier LAN Architecture
2. Three-Tier LAN Architecture
3. Spine-Leaf Architecture
4. SOHO (Small Office/Home Office) Networks

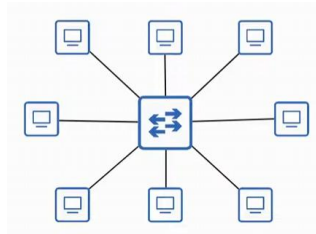
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Common Network Topologies

Before studying LAN architectures, three important topology terms are introduced.

1. Star Topology

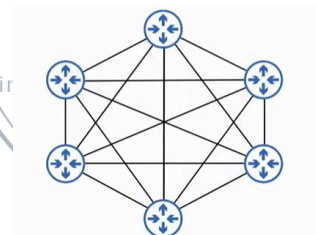
A star topology consists of multiple devices connected to one central device.



- All devices connect to a central device.
- The diagram does not need to look like a star.
- If multiple devices connect to a single central device, it is considered a star topology.

2. Full Mesh Topology

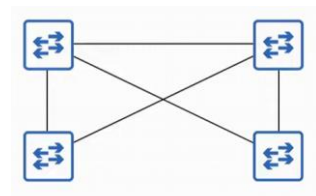
A full mesh topology exists when every device has a direct connection to every other device.



- Every device is directly connected to every other device.
- Example: six routers with direct links between all routers.
- Provides maximum redundancy

3. Partial Mesh Topology

A partial mesh topology exists when some devices have direct connections, but not all devices.



- Some devices are directly connected.
- Not every device has a direct link to every other device.

2-Tier and 3-Tier LAN Architecture

Two-Tier LAN Architecture

A Two-Tier LAN consists of two hierarchical layers:

1. Access Layer
2. Distribution Layer

This design is also called a **Collapsed Core Design** because the functions of the Core Layer are combined into the Distribution Layer.

Access Layer

The Access Layer is where end devices connect to the network.

Connected Devices

- PCs
- Printers
- IP Phones
- Security Cameras
- Wireless Access Points

Functions

1. End-Host Connectivity

Access switches provide the ports used by end devices.

2. QoS Marking

Traffic is typically marked at the access layer.

Reason: Traffic should be classified and marked as early as possible in the network.

3. Security Services

Common security functions include: Port Security & Dynamic ARP Inspection (DAI)

4. Power over Ethernet (PoE)

Access switches may provide PoE for:

- Wireless Access Points
- IP Phones

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Distribution Layer

The Distribution Layer aggregates connections from Access Layer switches.

Functions

1. Aggregation

Multiple access switches connect to distribution switches.

2. Layer 2 / Layer 3 Boundary

The distribution layer is typically the transition point between:

- Layer 2 Switching
- Layer 3 Routing

3. Protocol Operation

Distribution switches commonly run:

1. Layer 2 Protocols-STP
2. Layer 3 Protocols-OSPF

4. Default Gateway Services

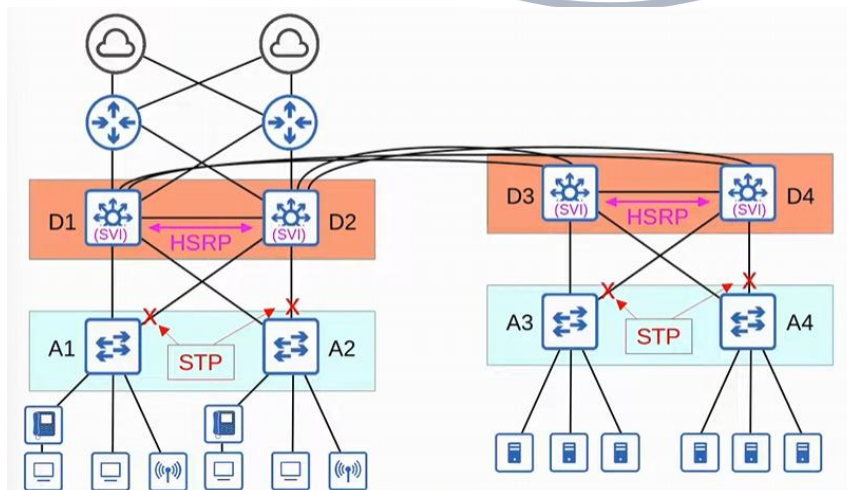
Hosts typically use SVIs configured on distribution switches as their default gateways.

5. External Connectivity

The distribution layer connects the LAN to:

1. Internet
2. WAN
3. Other LAN segments

Example Two-Tier Campus LAN



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Components

Access Layer

1. A1
2. A2

These switches connect end devices and wireless access points.

Distribution Layer

1. D1
2. D2

These switches aggregate access-layer connections.

Redundancy

Each access switch connects to both distribution switches.

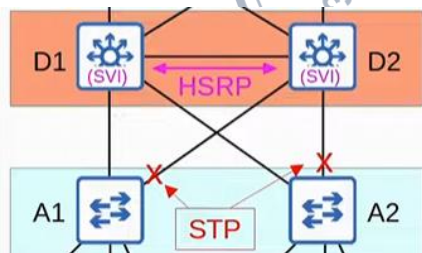
Benefits

- Alternative path available if one link fails.
- Improved network availability.

Spanning Tree Protocol (STP)

Because the access-to-distribution links are Layer 2 connections, loops can occur.

STP prevents loops by blocking certain redundant links.



Some links blocked by STP

Without STP, Layer 2 loops would occur.

Default Gateway Redundancy

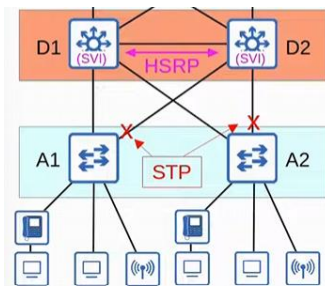
The distribution switches are multilayer switches.

Hosts use their SVIs as default gateways.

To provide gateway redundancy, FHRPs are used, such as:

- HSRP
- VRRP

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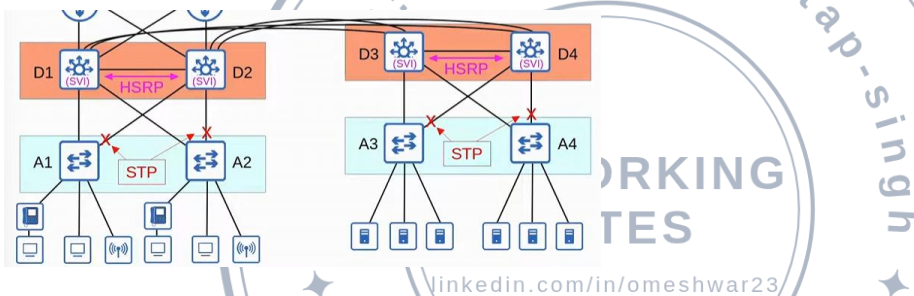
A virtual IP address is presented to hosts as the default gateway.

Connectivity to External Services

The Distribution Layer provides connectivity to:

1. Internet
2. WAN
3. Other sections of the LAN

Example



Distribution Switch Interconnections

In a Collapsed-Core Design, distribution switches connect to one another using Layer 3 links.

Important Points

1. No STP on these links.
2. Routing information can be exchanged using OSPF.
3. Distribution layer performs both distribution and core functions.

For this reason, it is sometimes called the: Core-Distribution Layer

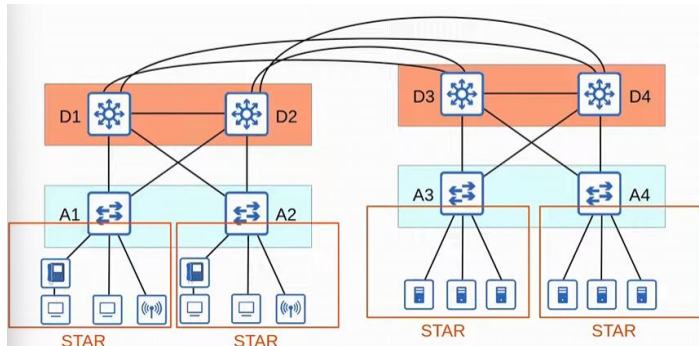


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Topologies Inside the Two-Tier Design

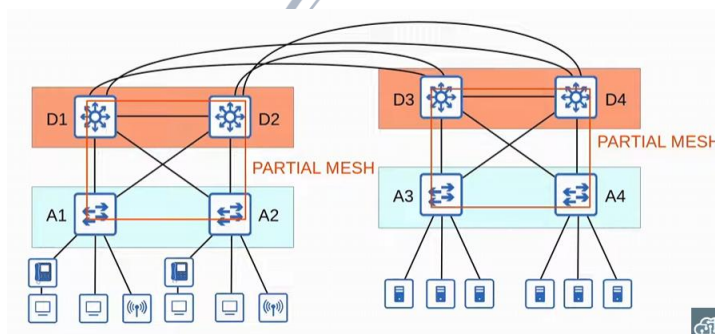
Star Topology

Each access switch forms a star topology with connected end devices.



Partial Mesh Topology

Between Access and Distribution Layers.

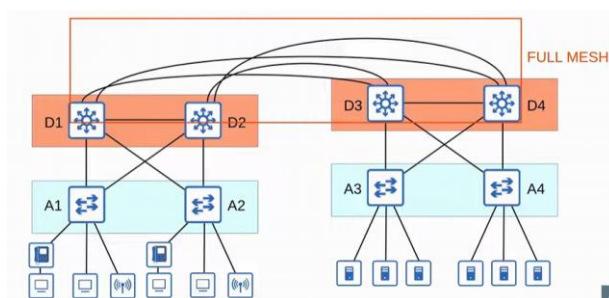


- Distribution switches connect to each other.
- Distribution switches connect to both access switches.
- Access switches do not directly connect to each other.

Therefore, this is a partial mesh topology.

Full Mesh Topology

Between Distribution Switches.



Every distribution switch connects directly to every other distribution switch.

This forms a full mesh topology.

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Scaling Problem

As the campus network grows, additional distribution layers may be added for different buildings or areas.

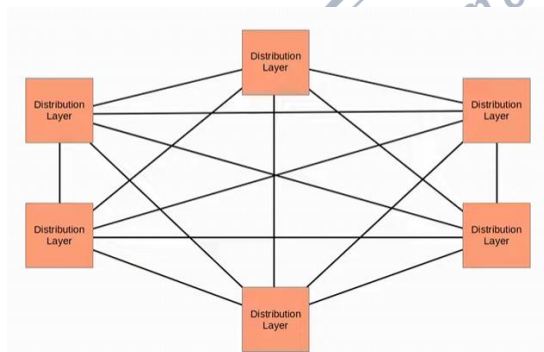
Challenge

The number of direct connections between distribution switches increases rapidly.

Result

- More cabling
- More complexity
- More difficult management
- Reduced scalability

Example



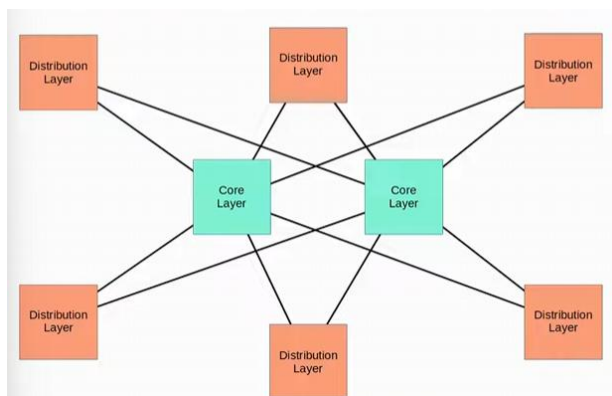
Large numbers of distribution switches require many interconnections.

Core Layer

To improve scalability, a Core Layer can be added.

Cisco Recommendation

Add a Core Layer when there are: More than three Distribution Layers in a single location.



Three-Tier LAN Architecture

The Three-Tier Architecture consists of:

1. Access Layer
2. Distribution Layer
3. Core Layer

Core Layer

The Core Layer connects separate Distribution Layers together in large LAN networks.

Its primary goal is: Fast transport of traffic across the network.

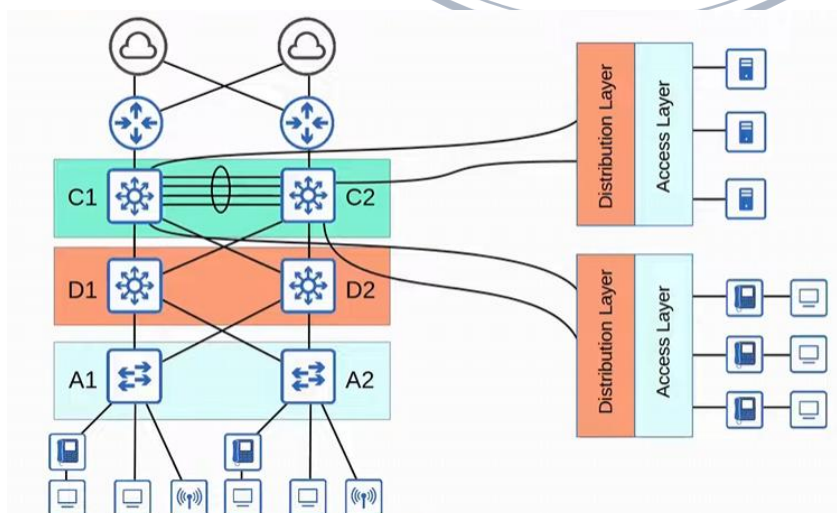
Because speed is the priority:

1. CPU-intensive operations should be avoided.
2. Security features should not be performed here.
3. QoS marking and classification should not be performed here.
4. The focus is fast packet forwarding only.

Core Layer Characteristics

1. Connects Distribution Layers together
2. Focuses on speed ("fast transport")
3. Uses Layer 3 connections only
4. Does not run Spanning Tree
5. Maintains LAN connectivity even when devices fail
6. Requires redundant devices and connections because it is the backbone of the LAN

Three-Tier Campus LAN Example



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In this design:

1. Core switches connect to Internet routers.
2. Additional Distribution and Access Layers also connect to the Core Layer.
3. Small LANs generally use two tiers.
4. Large LANs commonly use three tiers.

Layer Summary

Access Layer:

- the layer that end hosts connect to (PCs, printers, cameras, etc.)
- typically Access Layer Switches have lots of ports for end hosts to connect to
- QoS marking is typically done here
- Security services like port security, DAI, etc are typically performed here
- switchports might be PoE-enabled for wireless APs, IP phones, etc.

Distribution Layer: *sometimes called Aggregation Layer

- aggregates connections from the Access Layer Switches
- typically is the border between Layer 2 and Layer 3
- connects to services such as Internet, WAN, etc. *in a two-tier design

Core Layer:

- Connects Distribution Layers together in large LAN networks
- The focus is speed ('fast transport')
- CPU-intensive operations such as security, QoS marking/classification, etc. should be avoided at this Layer
- Connections are all Layer 3. No spanning-tree!
- Should maintain connectivity throughout the LAN even if devices fail

Important Note

Although two-tier and three-tier architectures are common and recommended by Cisco, network design does not have universal answers.³

Different enterprises may implement different variations depending on their requirements.

Spine-Leaf Architecture

Introduction To Spine-Leaf Architecture

Spine-Leaf Architecture was designed for **Data Center environments**.

Data Center

A Data Center is a dedicated space or building used to store:

1. Servers
2. Network Devices
3. Computer Systems

Data centers typically contain rows of racks where servers and networking equipment are mounted.

Traditional Data Center Design

Traditionally, data centers used the same:

1. Access Layer
2. Distribution Layer
3. Core Layer

architecture used in campus networks.

This design worked well when most traffic was: North-South Traffic

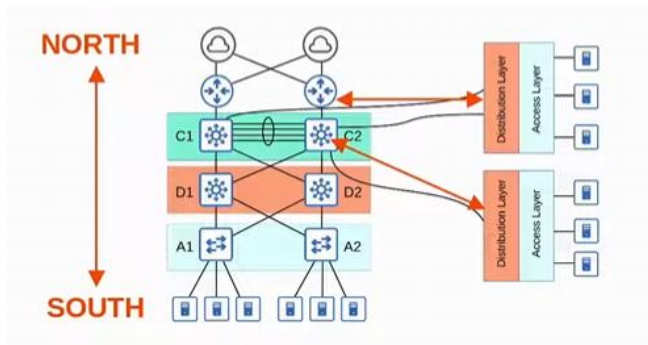
North-South Traffic

North-South traffic moves between:

- Access Layer
- Distribution Layer
- Core Layer
- Internet

or between different sections of the LAN through those layers.

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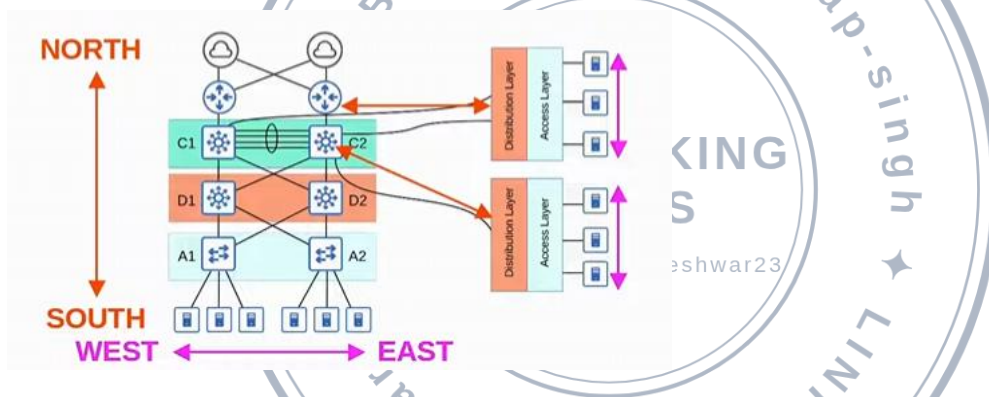
Traffic moving up and down the hierarchy is called North-South traffic.

East-West Traffic

East-West traffic is traffic exchanged directly between servers.

It remains inside the data center and does not travel to:

- Internet
- Other external network locations



Increase of East-West Traffic

With the growth of virtual servers and distributed applications:

- Applications are deployed across multiple physical servers.
- East-West traffic increases significantly.

As East-West traffic increased, the traditional three-tier architecture created:

- Bandwidth bottlenecks
- Variable server-to-server latency

The latency depended on the path traffic followed through the network.

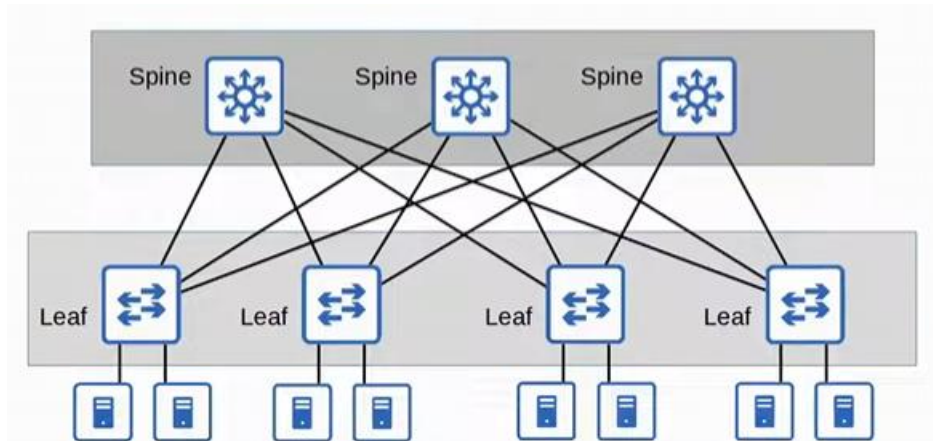
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Spine-Leaf Architecture

To solve these issues, **Spine-Leaf Architecture** became popular in data centers.

It is also known as: Clos Architecture

Spine-Leaf Structure



Rules of Spine-Leaf Architecture

1. Every Leaf Switch Connects to Every Spine Switch

Each Leaf switch has an uplink to every Spine switch.

2. Every Spine Switch Connects to Every Leaf Switch

This is the reverse view of the same rule.

3. Leaf Switches Do Not Connect to Other Leaf Switches

4. Spine Switches Do Not Connect to Other Spine Switches

5. End Hosts Connect Only to Leaf Switches

Servers attach to Leaf switches.

Leaf switches function similarly to the Access Layer.

Traffic Flow in Spine-Leaf

Traffic paths are selected randomly to distribute traffic load among Spine switches.

This balances traffic across the network.

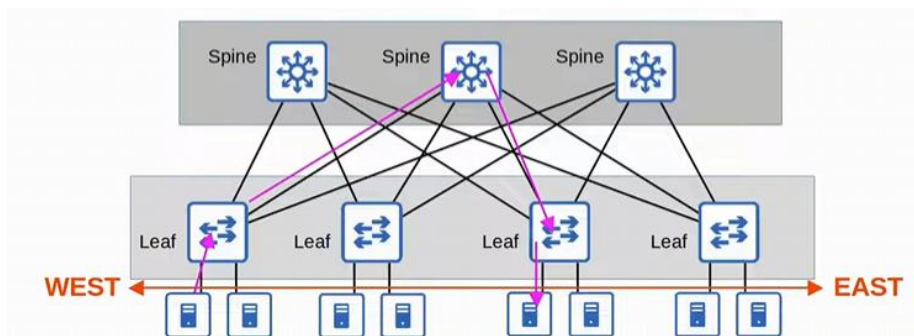
Consistent Latency

A major benefit of Spine-Leaf architecture is that servers are separated by the same number of hops.

The exception is servers connected to the same Leaf switch.

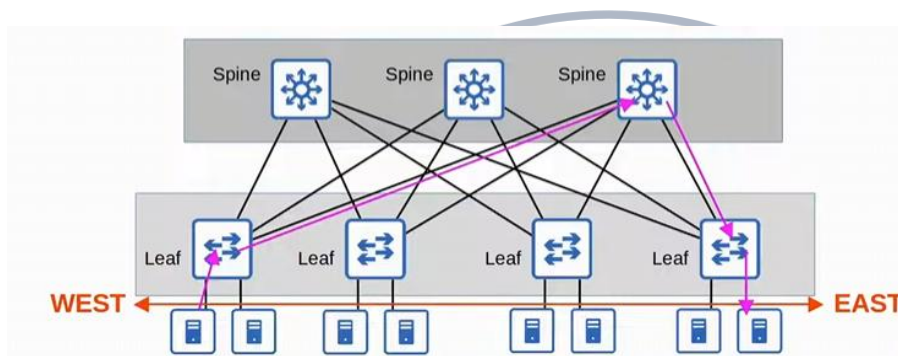
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Example 1



Switches traverse= 3 switches

Example 2



Again = 3 switches

Because server-to-server paths traverse the same number of switches, East-West traffic experiences consistent latency.

Scalability

Spine-Leaf architecture is easy to scale.

When more servers are needed:

1. Add another Leaf switch.
2. Connect it to all existing Spine switches.

This allows the network to grow while maintaining the same architectural principles.

SOHO Networks

Introduction

The final topic is SOHO Networks.

SOHO stands for: **Small Office / Home Office**

It refers to:

- The office of a small company, or
- A small home office with only a few devices

However, it does not actually need to be used as an office.

If a home has a network connected to the Internet, it is considered a **SOHO network**.

Characteristics of SOHO Networks

SOHO networks typically have simple networking requirements.

Because of these simple requirements, networking functions are usually provided by a **single device**.

This device is commonly called:

- Home Router
- Wireless Router

Unlike enterprise networks, there is no need for a separate device for each networking function.

Functions of a Home Router

A single home router can perform multiple networking functions.

1. Router

The device acts as a router by: Connecting the home network to the Internet.

2. Switch

The device also acts as a switch.

Typically:

- Several switchports are available on the back of the device.
- Other devices can connect directly to these ports.

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3. Firewall

Simple firewall functions are usually included.

Typically:

- Connections coming from the outside network are blocked.
- Connections initiated by internal devices toward the Internet are allowed.

4. Wireless Access Point

The device also functions as a wireless access point.

This allows devices such as:

- Laptops
- Phones

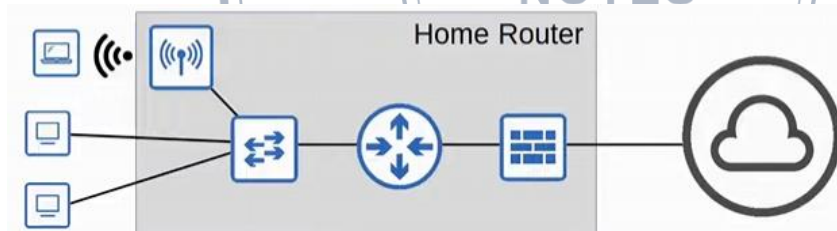
to connect to the network using Wi-Fi.

5. Modem (In Some Cases)

Sometimes the home router can also function as a modem for cable Internet.

In other cases: The modem is a separate device.

Example SOHO Network



The diagram shows that multiple networking functions are combined into a single device.

Why SOHO Networks Use One Device

For very small networks:

1. There is no need for a dedicated router.
2. There is no need for a dedicated switch.
3. There is no need for a dedicated wireless access point.
4. There is no need for separate devices for each networking function.

Instead: One device performs all of these functions.

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SOHO vs Enterprise Networks

In an enterprise network:

- Routing functions are typically provided by dedicated devices.
- Switching functions are typically provided by dedicated devices.
- Wireless access is typically provided by dedicated devices.
- Security functions are typically provided by dedicated devices.

In a SOHO network:

The home router combines all of these functions into a single small device.

